



The value of U.S. patents by owner and patent characteristics

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ABSTRACT

This paper uses renewal data to estimate the value of U.S. patents, controlling for patent and owner characteristics. Estimates of U.S. patent value are substantially larger than estimates for European patents, however, the ratio of U.S. patent value to R&D for firms is only about 3%. Contrary to a common assertion, patents issued to small patentees are much less valuable than those issued to large corporations. Litigated patents are more valuable, as are highly cited patents. However, patent citations explain little variance in value, suggesting limits to their use as a measure of patent quality.

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Patent value is an interesting economic quantity for several reasons: it informs policy because it is a measure of the reward that the patent system provides inventors; it aids accounting for the value of intangibles; it helps measure the productivity and quality of R&D.

This paper extends the research on patent value by estimating the renewal value of U.S. patents using a rich set of control variables including patent citation statistics, whether the patent was litigated or reissued, the type of patent assignee and technology, and, for a sub-sample of patents issued to publicly listed firms, details of the patent owner's financial and other characteristics, patent portfolio size, R&D, and industry. This extension allows quantitative assessment of the association between patent citations and patent value, more accurate estimation of the subsidy that patents provide to perform R&D, and better understanding of variation in patent value across different types of inventors.

This approach combines two strands of the literature. One strand uses data on patent renewal decisions to estimate the value of holding a patent.¹ The implicit value of a patent is revealed when its owner pays a renewal fee, implying that the patent is worth more than the fee required to keep it in force. Subject to some important assumptions, this approach has been used to obtain dollar estimates of patents. However, most of this research only looks at the aggregate value of patents. Schankerman (1998) and Lanjouw (1998) look at patent value by technology category and nationality of the patentee for French and German patents, respectively. But many other factors that might reasonably influence patent value have not been explored in this context.

The other strand of the literature looks at the relationship between patent value and a variety of patent

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¹ See Lanjouw et al. (1998) for a review of this literature. Recent additions include Baudry and Dumont (2006), Gustafsson (2005) and Serrano (2006).

characteristics with an eye to measuring patent “quality,” to developing quality-adjusted measures of inventive output and to estimating the contribution of intangibles to firm value. These studies look at correlations between patent characteristics and variables that should be correlated with patent value such as whether a patent is litigated or opposed (Harhoff et al., 2003b; Allison et al., 2004; Lanjouw and Schankerman, 2004a; Marco, 2005), survey measures of subjective value (Harhoff et al., 1999, 2003a), the number of countries in which the patentee files (Putnam, 1996; Lanjouw and Schankerman, 2004a), whether the patent is renewed (Harhoff et al., 2003b; Lanjouw and Schankerman, 2004a), and firm market value (Hall et al., 2005). Based on such correlations, researchers infer, for example, that the number of citations made to a patent is associated with that patent’s value. However, these studies recognize that the relationship between citations and patent value is “noisy” and they have not quantified the actual increase in value associated with an incremental citation received.

This paper integrates these two approaches. Because a rich set of data is available about U.S. patents, I can combine information about patent renewals with information about the owner and patent characteristics. I model patentees’ decisions to renew as a kind of ordered probit—patent renewal fees increase sharply with the age of the patent, sorting the patents by value over time. Variables such as patent citations and firm characteristics can be included on the right-hand side of the corresponding regression. The latent variable that represents the patentee’s valuation is a linear combination of such characteristics and a stochastic error term. In effect, previous studies using renewal data have only included a constant and an error term on the right-hand side.²

There are several advantages to including patent characteristics and owner characteristics in a renewal model of patent value. Using this model, I am able to obtain dollar estimates of the incremental effect of patent citations and other characteristics on patent value. I am also able to estimate how much of the total variance in log patent value is explained by such characteristics. I find, in general, that these characteristics are significantly correlated with patent value but they do not explain much of the overall variation. That is, as other researchers have concluded, these are rather noisy measures of value. Citation statistics may be more informative about the value of the underlying technology than they are about the value of the patent *per se*.

In addition, this approach allows me to measure separately other factors that influence patent value such as the type and firm size of the patentee or the size of the patentee’s patent portfolio. I find substantial differences in patent value, for example, between small and large patentees. This has important implications for policy.

This finer-grained information about patent owners also allows me to obtain more accurate estimates of the value that patents provide as incentives to perform R&D. Schankerman (1998) argues that the ratio of patent value

to the value of the associated R&D is an “equivalent subsidy rate”—under some assumptions, this ratio represents an upper bound on the subsidy that would be needed to elicit the same level of R&D in a world without patent protection. Several patent renewal studies have estimated this ratio either for the aggregate patents of a nation or by technology class, using highly aggregated data. Because I can closely match the value of patents to the value of R&D on a firm-by-firm basis, I am able to obtain more representative estimates of the equivalent subsidy rate.

Patent renewal data has been used previously to obtain estimates of the value of U.S. patents by Serrano (2006), who obtains preliminary estimates of the value of a subset of U.S. patents using patent renewal and re-assignment data, and by Barney (2002), who reports few details. Moore (2005) reports general features of U.S. patent renewal data.

The paper is organized as follows: the next section presents a model of patent renewal and discusses theoretical issues; Section 2 describes the data; Section 3 presents results for patents issued in 1991 and for a panel of patents issued to public manufacturing firms from 1985 to 1991. Section 4 compares my estimates with those obtained by other researchers, including estimates relative to R&D spending. Section 5 concludes.

1. Model of patent renewal and value

1.1. Patent value

Researchers have used measures of patent value for a variety of different purposes, including measuring inventive output, measuring the incentive effect of patents, and measuring the contribution of intangibles to firm value. Different uses can imply different definitions and different methods of measurement, so it is helpful to set out some initial distinctions (see Harhoff et al., 2003b for a different set of distinctions).

I aim to estimate the value of incremental rents that patents earn. Patents can provide their owners a degree of market power that conveys a stream of profits that exceeds the profits they could earn without patents. These profits can be realized either through the ability to exclude others from product markets, in which case product prices rise above the level without patents, or through the ability to exclude others in the markets for technology licensing or sale. This notion of patent value corresponds to the “reward” theory of patents—patent rents are the reward.

The value of patent rents, however, is distinct from the value of the underlying technology. This divergence occurs for two reasons. First, innovators also appropriate value from technology by non-patent means such as lead-time advantage and trade secrecy. The value of patent rents is incremental, that is, it is measured relative to an alternative value appropriated by these other means. In general, a patent protects neither all of the inventions nor all of the technical knowledge of a new technology, so in general, the value of a technology exceeds the value of the associated patents.

Second, the value of patents is, to some extent, endogenous. Patentees can exert varying degrees of effort in the

² Other studies using other data, such as Putnam (1996), have used richer sets of right-hand side variables.

prosecution of patents and in their enforcement. This effort at patent “refinement” affects the strength of the patent rights and hence the value of the rents derived. For instance, patent applicants can invest more effort in drafting a patent by including more claims (to broaden the scope of the claims and to make them more resistant to invalidation challenges) and more citations (to immunize the patent against possible prior art). Patent owners can strengthen patent claims by obtaining a re-issued patent. Innovators can also obtain more patents on a technology and related technologies. A larger number of patents may simply reflect that an owner is protecting more inventions with patents, or that owner may be “fencing” off the technology by patenting possible substitute technologies or building a patent “thicket.” Below I find evidence that patentees do increase aggregate patent value through such measures.

This endogeneity means that variation in the value of technologies does not necessarily correspond closely to the variation in patent rents. A firm with highly valuable technology may obtain relatively *more* patents on that technology, so that the average value of rents *per patent* does not entirely reflect the value of the technology. This means that patent value, in the sense used in this paper, does not serve well as a measure of “inventive output.” It will, in general, be less than the value of the underlying technology, although these two measures are likely correlated.

1.2. Patent renewal

The model I use is a simple variant of the model initially developed by Pakes and Schankerman (1984). Patentees derive rents from their patents only so long as those patents remain in force. If the expected stream of rents is not larger than the fees required to keep the patent in force, patent owners will let the patent expire. This means that patent renewal and expiration decisions implicitly reflect the value of the associated rents. Let $r_i(t)$ be the annual flow of rents for the i th patent at time t .

A renewal model requires two key assumptions about r : how it is initially distributed and how it changes over time. First, consider how the flow of rents changes over time. The initial renewal studies used a constant deterministic depreciation rate. A patent might depreciate because of technological obsolescence (the underlying invention becomes less valuable) or because competitors are able to “invent around” the patent. But patent holders might also learn about the value of their patents over time, so their expected profits might actually increase over time. Several papers have studied richer models where patents have an option value that evolves stochastically over time as new information is gained (Pakes, 1986; Lanjouw, 1998; Baudry and Dumont, 2006). However, these considerations do not appear to make much difference to the results. Estimates from these models are quite similar to those obtained from the same data using deterministic depreciation. And although Pakes (1986) and Lanjouw (1998) find evidence of an initial “learning” effect that increases the value of some patents, they also find that this effect is largely complete by the end of the fourth year after issue. After that, the profit flows decrease steadily.

These studies use European data, which has annual observations because European nations require annual payment of maintenance fees. The U.S. only requires three payments of maintenance fees, so U.S. data has far fewer degrees of freedom. The limited degrees of freedom of U.S. data do not permit me to estimate a model with a stochastic learning effect.³ Since my first observation occurs at the end of the fourth year – when any learning appears to be complete – this would not seem to be a problem. In the worst case, it might introduce a mild upward bias to value estimates. Consequently, following much of the literature, I assume that the profit flow depreciates at a constant rate, d , so that $r_i(t) = r_i(0)e^{-dt}$. To test for year-specific shocks, I run one set of regressions over panel data with dummies for different cohorts.⁴ Also, below I consider the possibility that some groups of patentees, such as foreign patentees, might experience a different time pattern of depreciation.

The second choice concerns the distribution. Researchers find that a lognormal generally fits the distribution of invention values well (Harhoff et al., 2003a) and patent renewal data are consistent with a lognormal distribution as well (Pakes and Schankerman, 1984). Several researchers have tested the lognormal, Pareto and Weibull distributions with renewal data and they have all found that the lognormal distribution provides the closest fit (Schankerman and Pakes, 1986; Lanjouw et al., 1998; Schankerman, 1998). These tests were performed using the richer European data; the limited degrees of freedom in the U.S. data do not permit comparably accurate tests of fit.⁵ In what follows, I use the lognormal distribution.

Note that the choice of distribution affects some inferences one may want to draw from the renewal model. The most valuable patents – the “upper tail” of the distribution – are all renewed to full term, so observations about renewal do not directly reflect the values of these patents. In effect, this means that although the distribution fits the majority of less valuable patents, statistics that reflect the upper tail are based on an extrapolation. For example, renewal estimates of median patent value are accurately captured since the median patent is not renewed to term, but estimates of *mean* patent value reflect the upper tail and are thus based on extrapolation. This means that estimates of mean value are sensitive to how “fat” the upper tail is. In the worst case, as with the Pareto distribution, the tail may be so fat that there is no finite mean.

In a related paper (Bessen, 2006a), I show evidence that extrapolations based on a lognormal distribution is not too far off the mark. First, using estimates of the value of firm’s patent portfolios based on firm market value regressions, I

³ Ignoring patent characteristics and some minor variance in fees, I have three equations (one for each renewal fee) in three parameters (two distribution parameters and a depreciation rate), leaving my model just identified.

⁴ For instance, Schankerman (1998) finds evidence of oil shocks in French patent renewal data.

⁵ I tested the lognormal, Pareto and Weibull distributions on US data and I could only get the maximum likelihood estimation routine to converge for the lognormal distribution. These tests used Stata’s maximizing routines. Perhaps with more effort convergence could be achieved, however, the lack of convergence itself suggests a poorly fitting distribution.

find evidence of rapid convergence to a finite mean. Second, I find that estimates of patent value derived in this paper are quite comparable to estimates derived from market value regressions in most industries⁶ and these estimates also correspond roughly to some benchmarks of known patent value. Nevertheless, it might still be the case some inferences are sensitive to the extrapolation to the upper tail. Below I consider this possibility in evaluating the relationship between patent citations and patent value.

In addition to the assumed lognormal distribution and constant depreciation, profit flow may also be a function of observable characteristics of the patent or of the patent's owner. Let X_i be a vector of such characteristics such that, in keeping with the assumption of a lognormal distribution:

$$\ln r_i(0) = \beta \cdot X_i + \varepsilon_i, \quad (1)$$

where ε is a normally distributed stochastic error with zero mean and standard deviation σ such that $\ln r_i(0) = N(\beta \cdot X_i, \sigma)$. In most of the renewal literature, X_i is treated as a simple constant. In this case, three parameters are estimated, d , σ , and the constant mean, and these are sufficient to determine the median and mean values of patents. However, my data permit X_i to also include characteristics of the patents, such as citations received, and characteristics of the owner. Then estimates permit calculation of the direct dollar value effect of citations and these other characteristics. Putnam (1996), Harhoff et al. (2003b) and Deng (2005) also use a multivariate approach in slightly different contexts.

1.3. Estimation

The observations in the data set concern patentees' decisions whether to pay renewal fees to keep their patents in force for additional time periods. Patents applied for on or after 11 December 1980, accrue fees after 3.5, 7.5 and 11.5 years in order to remain in force beyond 4, 8 and 12 years, respectively. The fee schedules vary over time and they also depend on whether the assignee has "small entity" status—small firms, individuals and non-profit organizations pay fees that are only half of those paid by "large entities."

As in Europe, the fees increase with the age of the patent, that is, 12th year fees are much larger than 4th year fees. This is important because it means that the patentees' optimal renewal decision rule need only consider the current renewal period. That is, it is straightforward to show that if a patentee finds it unprofitable to renew at year 4, it will be even more unprofitable to renew at years 8 and 12, so an optimal rule will be to renew if and only if it is profitable to renew for the next immediate period. Specifically, if the payment of the fee keeps the patent in force for T more years at time t , then the patentee will renew the i th patent if and only if the present value of profits during those T years exceeds the renewal fee, c_{it} .

The present value of profits from t to $t+T$, is

$$\int_t^{t+T} r_i(\tau) e^{-s\tau} d\tau = r_i(0) z_t, \quad (2)$$

where $z_t = e^{-dt} \frac{1 - e^{-(d+s)T}}{d+s}$,

and s is the discount rate. I follow the literature and use a discount rate of 10% per annum. Then the decision rule is to renew if and only if

$$\ln r_i(0) \geq \ln \left(\frac{c_{it}}{z_t} \right). \quad (3)$$

Given the lognormal distribution and substituting (1) into (3), the probabilities that a patent will expire at each given year are

$$\begin{aligned} P[\text{patent } i \text{ expires at } 4] &= \Phi \left(\frac{\ln(c_{i4}/z_4) - \beta \cdot X_i}{\sigma} \right), \\ P[\text{patent } i \text{ expires at } 8] &= \left[1 - \Phi \left(\frac{\ln(c_{i4}/z_4) - \beta \cdot X_i}{\sigma} \right) \right] \\ &\quad \Phi \left(\frac{\ln(c_{i8}/z_8) - \beta \cdot X_i}{\sigma} \right), \\ P[\text{patent } i \text{ expires at } 12] &= \left[1 - \Phi \left(\frac{\ln(c_{i8}/z_8) - \beta \cdot X_i}{\sigma} \right) \right] \\ &\quad \Phi \left(\frac{\ln(c_{i12}/z_{12}) - \beta \cdot X_i}{\sigma} \right), \\ P[\text{patent } i \text{ expires at } 17] &= 1 - \Phi \left(\frac{\ln(c_{i12}/z_{12}) - \beta \cdot X_i}{\sigma} \right), \end{aligned} \quad (4)$$

where Φ is the cumulative standard normal distribution function. This structure is, in fact, the same as that of an ordered probit, with the additional estimation of d (z is a function of d ; also, σ here provides more than just a scaling constant). I estimate this model by maximizing the likelihood function implied by (4). That is, each patent's contribution to the log likelihood function is the log of the right-hand side given in (4) corresponding to that patent's expiration date.

1.4. Calculating patent value

To estimate the net present value of patents, I perform a Monte Carlo simulation using the actual data. With estimates of the parameters, $\hat{\beta}$, $\hat{\sigma}$, and $\hat{d}$, in hand, I first determine bounds on ε_i for the i th observation conditional on observed renewal decisions made for that patent. For example, if the i th patent was allowed to expire after 8 years, then

$$\ln \left(\frac{c_{i4}}{z_4(\hat{d})} \right) - \hat{\beta} \cdot X_i \leq \varepsilon_i \leq \ln \left(\frac{c_{i8}}{z_8(\hat{d})} \right) - \hat{\beta} \cdot X_i. \quad (5)$$

For each observation in each Monte Carlo iteration, I select ε_i as a random draw from the lognormal distribution determined by $\hat{\beta}$ and $\hat{\sigma}$, conditional on (5) (or the corresponding bounds for patents with different expiration dates). Given this random draw, I calculate the corresponding $\ln r_i(0)$, and, from this, the present value of the patent at the time of issuance net of the discounted value of renewal

⁶ The main exception is for large pharmaceutical firms, where renewal estimates might be understated.

fees that will be paid. I repeat the Monte Carlo iterations a sufficient number of times so that the total number of observations exceeds 500,000.

2. Data

2.1. Samples

This study uses two datasets. The first consists of almost all U.S. utility patents issued in 1991 (patents assigned to governments and foreign individuals are excluded). The second is a panel of patents issued from 1985 through 1991 and assigned to publicly listed, R&D-performing firms whose primary line of business is in a manufacturing industry.

I used 1991 as the terminal year because this is the last year for which the final patent renewal decision could be observed (in 2003) when this research began.⁷ Patents applied for before 11 December 1980 are exempt from renewal fees. Because of this 1985 was the earliest year with few such exempt patents.

I obtained patent expiration data from the website of the U.S. Patent and Trademark Office (PTO).⁸ This data also included the patent's "entity" status at the time each fee was paid. The PTO designates individuals, small businesses and non-profit organizations as "small" entities. For most patents, the renewal fees for small entities were half those for large entities. In addition, the fee schedule was changed regularly and the fee for any given patent also depended on its issue date and, in some cases, on its application date.⁹ Some patents were reissued—a procedure where the patent owner can modify patent claims. I tracked reissues through to their final expiration as well. A small number of patents are reported as having missed a payment, but have petitioned to have the patent reinstated. I record these as if the last reported payment had been made.

Of the 96,513 patents issued in 1991, I obtained patent expiration information on 94,343. I excluded 1962 patents issued to governments or foreign individuals. Also, 33 patents were applied for before 11 December 1980 and were thus exempt from renewal fees. This left 175 patents that I could not find in the PTO's database.

In addition to this data for 1991, I assembled a panel of firm data to explore additional variables and to explore the regressions over time. To this end, I assembled a panel of patents owned by publicly listed manufacturing firms from 1985 through 1991. I drew this panel dataset from a larger sample developed for another project (Bessen and Meurer, 2005) that matched patent data to firms in the Compustat dataset of firm financial information. The USPTO provides an assignee name for every assigned patent after 1969. To match the USPTO assignee name to the Compustat firm name, we began with the match file provided

by the NBER (Hall et al., 2001). To this we added matches on subsidiaries developed by Bessen and Hunt (2004), we manually matched names for large patenters and R&D performers, and we matched a large number of additional firms using a name-matching program. In addition, using data on mergers and acquisitions from SDC, we tracked patent assignees to their acquiring firms. Since a public firm may be acquired, yet still receive patents as a subsidiary of its acquirer, we matched patents assigned to an acquired entity in a given year to the firm that owned that entity in that year.¹⁰ The matched group of firms accounts for 96% of the R&D performed by all U.S. Compustat firms, 77% of all R&D-reporting firms listed in Compustat and 62% of all patents issued to domestic non-governmental organizations during the sample period. Sample statistics show that this sample is broadly representative of the entire Compustat sample, although it is slightly weighted toward larger and incumbent firms.

From this larger sample, I selected a panel of firms from 1985 through 1991 that had a primary line of business in a manufacturing industry, that performed R&D, and that had at least 4 years of non-missing data in key variables. This left me with a sample of 107,364 patents issued during this time period to 1066 firms.

3. Regression results

3.1. Patent value by groups

3.1.1. Renewal rates

Table 1 shows summary expiration and renewal data by major groups for the 1991 sample. Overall, only 41.52% of patents were renewed to full term, with roughly equal groups dropping out at each renewal stage. The mean renewal fees were not large, however, increasing from \$814 after 4 years, to \$1562 after 8 years, to a final mean payment of \$2327. About 29% of the patents were issued to patentees who were "small entities" at year 4. In most cases, these patentees pay fees that are half as large as those paid by large entities.

Among assignee types, patents that were unassigned at issue or issued to individuals had the lowest rate of renewal to term, 22%, while publicly listed U.S. firms had the highest rate, 50%. Relatively few publicly listed U.S. firms and foreign organizations rated as small entities. Similarly, patents awarded to small entities in general were much more likely to expire despite lower fees—only 25% were renewed to full term compared to 48% for patents held by large entities.

Table 1 also shows these variables for six technology categories defined by Hall et al. (2001), based on each patent's primary technology class as assigned by the PTO. Computer and communications patents had the highest rate of renewal to full term, while "other" technologies had the lowest rate. Drug and medical patents had the high-

⁷ Patentees who miss the deadline for paying the fees are given a grace period during which they can pay with a penalty. They can also appeal expirations arising from missed payments.

⁸ <https://ramps.uspto.gov/eram/patentMaintFees.do>.

⁹ I obtained the details of the fee schedule over time from the Federal Register and Public Laws. This information is available from the author.

¹⁰ This dynamic matching process is different from that used in the original NBER data set which statically matched a patent assignee to a Compustat firm. These data were developed with the help of Megan MacGarvie, to whom I am indebted.

Table 1
Summary statistics

	Percent expired during			Percent full term	Percent small	Number
	4th year	8th year	12th year			
All	20.21	20.95	17.31	41.52	29.33	94,342
Assignee type						
Unassigned and individuals	36.05	26.14	15.43	22.38	88.24	17,786
Non-public organizations	18.21	20.41	17.00	44.38	45.34	17,229
Publicly listed firms	13.70	19.37	16.58	50.35	9.77	21,904
Foreign organizations	17.43	19.66	18.78	44.13	2.34	37,423
PTO entity status						
Small	32.22	25.72	16.66	25.40	100.00	26,768
Large	15.46	19.06	17.57	47.91	0.00	67,574
Technology category						
Chemical	19.10	21.19	18.63	41.08	15.73	18,175
Computers and communications	11.74	17.46	17.56	53.24	14.51	9,816
Drugs and medical	20.11	20.66	15.13	44.10	36.87	8,288
Electrical and electronic	16.28	19.28	17.45	46.99	18.91	16,481
Mechanical	21.65	21.62	17.72	39.00	31.73	21,561
Other	27.11	23.22	16.34	33.33	51.82	20,021
Mean fee (\$92)	814	1562	2327			

est proportion of small entity patents, many of these from inventors making medical devices.

3.1.2. Regression results

Table 2, column 1, shows a basic regression for all U.S. patentees (excluding patents assigned to foreigners and governments). Because fees differ sharply depending on the patentee's entity status and because there may also be important differences between large and small entities, I control for small entity status in all regressions. Of course, entity status can change over time. Small firms grow and, more frequently, valuable patents owned by small patentees are acquired by large firms. These regressions include a dummy variable if the patentee was a small entity in year 4, and dummies if the entity status changed (up or down) from years 4 to 12. I also estimate σ and the depreciation rate and I report median and mean net present values for these patents calculated using the Monte Carlo technique described above.

The estimates for σ (1.86) and the depreciation rate (14%) are broadly similar to the corresponding parameters found in studies of European patents. However, the means are quite different, generating much higher patent values. I estimate a mean value for patents granted to U.S. patentees of \$78,168 and a median value of \$7175, also in 1992 U.S. dollars. In Section 4, below, I compare these estimates to those obtained by other researchers.

3.1.3. Patent value for small patentees

The coefficients on the entity size dummy variables suggest that patents owned by small entities are dramatically less valuable than patents owned by large entities. This is confirmed by separate regressions in columns 2 and 3 for patentees who were small and large entities in year 4, respectively, although the difference in the means is not as great as the fivefold difference in the median values or the implied profit flows. Also, the small number of patents that were owned in year 4 by a small entity but owned by a large entity in year 12 were substantially more valu-

able than most other patents.¹¹ This suggests a selection effect: the most valuable patents owned by small entities are acquired by large entities.

A similar selection effect may explain part of the reason small entity patents in year 4 are less valuable than large entity patents—some patents initially issued to small entities are acquired by large entities by year 4. I do not have data on the initial status of each patent. However, the aggregate numbers on small entity patents suggest that this selection effect is not large because relatively few patents are transferred during this interval. At issuance, 30.17% of 1991 patents were owned by small entities while 29.33% were at year 4.¹² Assuming, say, that patents transferred to large entities have a log profit flow that is 3.93 larger (column 2), then counting the patents that transferred would mean that the log profit flow of small entity patents was only 1.81 less than the log profit flow of large entities at issuance.

In simpler words, patents owned by individuals, small companies and non-profit organizations have much lower values than those owned by large companies even after taking into account a selection effect. This is important because it is sometimes argued that patents are particularly valuable to small patentees, since large firms may have more alternatives to patent protection, such as complementary products or services. These results suggest, instead, that patents do a relatively poorer job of earning returns for small inventors compared to large firms.

I explore this further in Table 3, which looks at regressions by assignee type. Individual assignees, including patents that were not assigned at issue (and therefore owned by the individual inventors by default), have the

¹¹ Among the 24,015 patents owned by small U.S. entities in year 4, 1309 were owned by large entities in year 12.

¹² The latter number comes from Table 1. The former is derived from data on 1991 issuance fees in Lehman (1993). Large entity issuance fees for utility patents collected were \$67,122 and small entity issuance fees, at half the rate, were \$29,004 (in thousands). $29,004 / (29,004 + 67,122/2) = 0.3017$.

Table 2
Regressions for U.S. patentees by size status

	1. All	2. Small entities	3. Large entities
Small entity, year 4	−1.92 (0.04)		
Entity change (small to large)	3.78 (0.11)	3.93 (0.14)	
Entity change (large to small)	0.54 (0.10)		0.54 (0.10)
Constant	8.45 (0.09)	6.72 (0.09)	8.37 (0.09)
σ	1.86 (0.05)	1.93 (0.07)	1.88 (0.06)
d	0.14 (0.01)	0.16 (0.01)	0.13 (0.01)
Median net present value	7174.6	2942.9	14310.2
Mean net present value	78167.7	70100.2	105916.2
Number of observations	56816	24015	32801
Log L	−72444.1	−31600.7	−40809.9

Note: Robust standard errors in parentheses. Entity change dummy variables reflect change in status between the fourth year and final year. Patent values in 1992 \$ and discounted at 10%.

Table 3
Regressions by assignee type

	U.S. assignee			4. Foreign organization	5. Foreign organization with $d = 0.15$
	1. Unassigned or individual	2. Non-listed organization	3. Publicly listed firm		
Small entity, year 4	−.156 (0.06)	−1.39 (0.06)	−1.27 (0.11)	−1.94 (0.07)	−1.54 (0.04)
Entity change (small to large)	3.91 (0.19)	3.53 (0.20)	2.89 (0.23)	4.55 (0.24)	3.34 (0.13)
Entity change (large to small)	0.83 (0.18)	0.64 (0.16)	0.92 (0.20)	1.61 (0.25)	0.89 (0.16)
Constant	7.87 (0.13)	8.71 (0.22)	8.38 (0.12)	9.93 (0.18)	8.43 (0.01)
σ	1.79 (0.08)	2.06 (0.13)	1.80 (0.07)	2.82 (0.11)	1.93 (0.01)
d	0.14 (0.01)	0.16 (0.02)	0.13 (0.01)	0.27 (0.01)	0.15–
Median net present value	2588.9	19206.3	49297.7	62326.5	16421.5
Mean net present value	25597.6	206286.3	90787.8	2905760.7	107906.3
Number of observations	17786	17229	21904	37423	37526
Log L	−23338.6	−21770.5	−26962.9	−48269.4	−48443.2

Note: Robust standard errors in parentheses. Column 5 fixes the depreciation rate at 15%. Entity change dummy variables reflect change in status between the fourth year and final year. Patent values in 1992 \$ and discounted at 10%.

lowest patent values, \$25,598 in the mean. Patent values from organizational inventors (mostly firms) are larger, as seen in columns 2 and 3. Interestingly, the distribution for non-public organizations appears to be more highly skewed than the distribution for public firms, generating a higher mean value despite a lower median value. But the lower values of patents owned by small entities is not just a matter of individual inventors—similar relative values are found across all types of assignees.

Thus the gap between the values of patents held by small patentees and those held by large firms is large, persistent and it holds for many different sorts of small patentees. This finding is consistent with limited additional evidence. [Arora et al. \(2003\)](#) also find a large disparity between large and small U.S. firms. [Gustafsson \(2005\)](#) finds a large disparity in patent value between individual and firm patents in Finland.

This result does contradict the common assertion that small patentees have particularly valuable patents. However, several points need to be considered in interpreting this result:

1. It does not imply that the inventions of small inventors are less valuable, just that small inventors are able to realize less value from their inventions through patents.

2. Individual inventors in low-tech fields might not have very valuable inventions—they have the lowest patent values. Nevertheless, other small inventors, such as small firms, might have inventions that are at least as valuable as the inventions of large firms. The small firms may simply realize relatively more value from their inventions by means other than patents such as lead-time advantage or because they occupy profitable niches, etc.
3. The gap in values might not exist in technology fields where markets for technology work well. Small firms may often lack the complementary assets to develop a technology and bring it to market. But with well-functioning technology markets, they can license or sell their technologies to firms that have the necessary assets, realizing comparable value. For example, [Hall and Ziedonis \(2001\)](#) find evidence that technology markets played an important role in facilitating the rise of “fabless” semiconductor design firms who outsource all manufacturing. I identified 111 patents issued to these semiconductor design firms in the 1985–1991 panel of public firms.¹³ Of these, 85% (94) were renewed to full term, suggesting that these were, indeed, highly valuable patents. Conversely, this suggests that the more general

¹³ Thanks to Rosemarie Ziedonis for sharing her data with me.

value disparity between large and small inventor patents may be an indication that technology markets do not work so well generally.

4. The gap probably cannot be attributed to higher monitoring and enforcement costs for small inventors. [Lanjouw and Schankerman \(2004b\)](#) argue that these costs are greater for small inventors. However, this does not seem to be an adequate explanation for the lower mean values of small entity patents for the following reason: faced with larger post-issue costs, small inventors will rationally only patent relatively more profitable inventions. Given a lognormal distribution of patent values, greater post-issue costs should actually *increase* the observed mean patent value even after taking those costs into account.¹⁴

3.1.4. Other patentee characteristics

Foreign organizations, which were not included in the previous regressions, appear to earn nearly \$3 million per patent (see column 4). Previous estimates based on renewal data have also reported exceptionally high patent value for Japanese patents and a majority of the foreign patents in this sample are from Japanese inventors.¹⁵ However, there may be a good reason why these estimates may not be reliable. The model assumes a constant rate of depreciation for the profit flow from a patent. It may well be that this assumption does not hold for foreign patenters. For example, foreign patenters may apply for a patent long before they are ready to market the invention in the United States. U.S. priority rules require that a U.S. patent be filed within a year after the invention is used or publicly disclosed, but typically a firm markets an invention first to its domestic market, only later rolling out sales and production to foreign markets. In this case, the pattern of profit flow for foreign patentees may diverge substantially from one with constant depreciation. There is some evidence that the estimate is off in this regard: the estimated depreciation rate in column 4 is nearly twice that of the other regressions. To control for this possibility, I repeat the same regression in column 5, but hold the depreciation rate constant at 15% per annum. With this change, the estimated mean net present value is \$107,906, just modestly larger than the mean present value estimate for U.S. public firms.

[Table 4](#) reports the results of separate regressions for different technology classes, similar to estimates for French technology classes by [Schankerman \(1998\)](#) and estimates for German technology classes by [Lanjouw \(1998\)](#). The technology categories I use were developed by [Hall et al. \(2001\)](#) and are based on the USPTO patent classification system. Contrary to the European studies, I find the highest mean patent values in chemicals and pharmaceuticals and the lowest values in computers, communications

Table 4

Estimates for technological categories for patents held by U.S. patentees

	Median net present value	Mean net present value
Chemical	33,856	497,200
Computers and communications	21,287	45,247
Drugs and Medical	12,692	120,419
Electrical and Electronic	11,928	68,459
Mechanical	8,171	86,033
Others	4,573	38,626

Note: Technology categories are from [Hall et al. \(2001\)](#). Patent values in 1992 \$ and discounted at 10%.

and “other.”¹⁶ Indeed, using the estimates in this table, the aggregate value of patents appears to be concentrated in chemical, drug and medical technologies: these technologies account for 70% of aggregate patent value but only 28% of the patents. Note also that although computers and communications patents have the second lowest mean value, they also have the second highest *median* value. This suggests that perhaps the mean values are low because there are just fewer “blockbuster” patents in this technology.

3.2. Patent value and patent characteristics

As noted above, many researchers have related patent characteristics to patent value. My model permits some of these associations to be quantified. [Table 5](#) includes patent characteristics in the regression. Column 1 shows characteristics that depend on choices made by the patentee. A patentee, aware that some patents are more valuable than others, may take efforts to make sure that the patent is more successfully enforced. These efforts at patent “refinement” include litigating, making more citations and claims in the patent application, and, possibly, re-issuing the patent. A re-issuance procedure permits a patentee to modify claim language, in some cases increasing the scope of the claim.

Each of these actions has a positive and statistically significant coefficient, suggesting that patent value is, to some extent, endogenous. Previous research has also found positive and significant relationships between patent claims and patent value as reported by survey respondents and between patent citations made and reported value ([Harhoff et al., 2003a](#)). Using the coefficients in the table, I can quantify all of these relationships. The last column of the table reports the percentage increase in profit flow associated with an incremental increase in the variable (e.g., one additional citation). A litigated patent is, all else equal, nearly six times more valuable. At the mean, a litigated patent is worth nearly half a million dollars. This corresponds well with what we know about litigation costs for plaintiffs, since patentees should only litigate those patents that are more valuable than litigation costs. In 1994, according to a survey of intellectual property lawyers (AIPLA 1994), the median legal cost of a patent lawsuit was \$190,000 through

¹⁴ With highly skewed distributions of patent value such as the lognormal distribution, an increase in a lower truncation threshold increases the mean value of the truncated distribution by more than the increase in the threshold, as long as the threshold is sufficiently large.

¹⁵ I also ran regressions by nationality, however, after Japan the numbers were too small to obtain reliable results.

¹⁶ Schankerman finds that pharmaceuticals and chemicals have the lowest mean values and he attributes this to price regulation for pharmaceuticals in France. Lanjouw finds a middling value for pharmaceutical patents.

Table 5
Regressions with patent characteristics

	1	2	3	4	Proportional increase (%)
Small entity, year 4	–1.89 (0.04)	–1.38 (0.04)	–1.84 (0.06)		
Entity change (small to large)	3.65 (0.11)	3.06 (0.12)	3.51 (0.15)	–1.82 (0.03)	
Entity change (large to small)	0.44 (0.10)	0.55 (0.12)	0.38 (0.11)	3.57 (0.10)	
Litigation	1.77 (0.132)	1.36 (0.172)		0.48 (0.09)	487
Reissued	1.02 (0.179)	0.84 (0.235)			179
Citations made	0.007 (0.001)		0.004 (0.001)		1
Self-citations		0.027 (0.005)			3
Citations to others' patents		0.000 (0.001)			0
Number of claims	0.017 (0.001)	0.006 (0.001)	0.013 (0.001)		2
Citations received		0.046 (0.003)	0.039 (0.003)	0.069 (0.003)	}4–7
Cites received ² /1000				–0.405 (0.048)	
Generality			0.173 (0.04)		
Originality			–0.106 (0.04)		
Constant	8.13 (0.09)	8.07 (0.11)	7.97 (0.14)	8.05 (0.07)	
σ	1.85 (0.05)	1.85 (0.07)	1.80 (0.09)	1.83 (0.05)	
d	0.14 (0.01)	0.14 (0.01)	0.14 (0.01)	0.14 (0.01)	
Number of observations	56816	38236	48990	56816	
Log L	–72014.8	–46816.4	–60891.0	–71331.4	

Note: Robust standard errors in parentheses. Litigation dummy is 1 for patents that were the main patent in one or more lawsuits filed by 1999 and listed in Derwent's Litalert service. Citation statistics are from the NBER patent database and are described in Hall et al. (2001). Entity change dummy variables reflect change in status between the fourth year and final year.

the discovery phase (after which many suits are settled) and \$301,000 through trial (costs have escalated substantially since then).¹⁷ A re-issued patent, all else equal, is nearly three times as valuable as other patents. Each additional citation made increases value about 1% and each claim increases value about 2%.

In column 2, I break out citations made to an assignee's own patents (self-citations) from those made to others' patents and I add patent citations received. It appears most of the value realized through citations made occurs from citations made to the patentee's own patents. This may be an indicator of "fencing" or "thicket building" behavior (Hall et al., 2005) where patentees strengthen their patents by also patenting related technologies or alternative technologies. Each self-cite increases patent value about 3%.

Since Trajtenberg (1990), researchers have used the number of citations that a patent receives as an indicator of patent or invention value. Previous research has found correlations between patent citations received and patent value reported in surveys (Harhoff et al., 2003a,b) and between patent citations received and firm market value (Hall et al., 2005). My coefficient on citations received is significantly associated with patent value and this statistic does seem to have greater statistical significance than the coefficients on citations made and claims. The economic significance of an additional citation received is also greater—an additional citation increases estimated profit flow by about 5% in this specification.

Column 3 drops the litigation and reissue dummies and adds statistics (calculated in the NBER database) for generality and originality. Generality and originality are measures suggested by Trajtenberg et al. (1997) that range from 0 to 1 and capture the technological diversity of citing and

cited patents, respectively. If the patents that cite the subject patent come from a large (small) number of technology classes, then generality will be high (low). If patent citations correspond to use of the technology in the cited patent, then high generality suggests that the cited invention is a general-purpose technology with many applications. Correspondingly, if a patent cites other patents from a large (small) number of technology classes, then it will have a high (low) originality index. Both of these measures have statistically significant coefficients, with a positive coefficient for generality and a negative one for originality.

Column 4 explores non-linearity in the effect of citations received by adding the square of this variable. The negative and significant coefficient on the squared term suggests diminishing returns to this effect. At the sample median (four citations received), an additional patent citation received increases profit flow by about 7% under this specification.

These results confirm general findings about the correlation between citation statistics and patent value in the literature. But my results also suggest that these associations have relatively small economic significance. At the sample mean, for example, an additional patent citation received corresponds to an increase in patent value of between three and five thousand dollars. This is substantially less than the effect suggested by some other research. For instance, Hall et al. (2005) estimate the relationship between firm market value and patent citations. Their results imply that at the sample mean, an additional citation received on a single patent corresponds to an increase in firm value of about \$327,000 (\$512,000 at the sample median).¹⁸ This large difference likely just means that we

¹⁷ These estimates are corroborated by a study of legal costs for patent lawsuits where judges order one party to pay the other's legal fees (Bessen and Meurer, 2007).

¹⁸ They report that an increase of one citation/patent for all the patents a firm owns increases firm market value by 2.7% at the mean. Mean market value is \$916.33m and mean patent stock is 75.72, yielding an increase of \$327,000 in market value with one citation on one patent.

are measuring different things. Hall et al., measure the relationship between a patent citation and the value of the technology to the firm generally; I measure specifically the effect of a patent citation on the value of the rents generated by a patent *per se*. As noted above, the value of the technology may be much greater than the value of the patent.

In addition, my estimates can be used to evaluate the portion of total variance in patent rents that can be “explained” by citation statistics. Given a vector of citation statistics, $\bar{X}$ which are a subset of the right hand variables, X , and given coefficients on these citation statistics of $\hat{\beta}$, the portion of variance accounted for by these statistics is

$$\frac{\text{var}(\hat{\beta} \cdot \bar{X})}{\text{var}(\hat{\beta} \cdot X) + \hat{\sigma}^2}.$$

I calculated this quantity for the various specifications in Table 5 for all of the citation statistics and just for citations received. In no case did the portion of variance explained equal as much as 6%. In other words, as other researchers have also concluded, patent citation statistics are correlated with patent value, but they are very “noisy signals.” This analysis indicates just how noisy they are.

I also measured this statistic across various subgroups: by technology category, by patent life and by large/small entity status. There was some variation, particularly among technology categories, where the portion of variance explained ranged from 2.2% for “Other” up to 12.0% for “Computers and communications.” The portion of variance explained was generally greater for technology groups that have higher levels of citations. Still, even in these cases, the portion of variation explained by patent citation statistics is small.

One possible weakness with this analysis is the possibility that citations may exert their greatest effect among those patents that are renewed to term. Since I do not directly observe values for this group of patents, my estimates may understate the explanatory power of patent citations. Several pieces of evidence suggest that this is not the case. First, the effect of patent citations on patent value is small *relative* to the effect of other patent characteristics. For example, litigation is associated with an increase in patent value that is over 100 times as much as the increase associated with a patent citation. Moreover, the value premium on litigated patents seems to be of the right order of magnitude based on the thumbnail calculations above. This suggests that possible “fatness” in the upper tail does not substantially bias estimates of the effect of patent characteristics. If estimates for litigated patents made in the upper tail appear to be roughly correct, why should estimates for highly cited patents be understated?

One possibility might be that citations increase not just the mean of the distribution, but they might increase the variance as well. Then the tail might be fatter for highly cited patents specifically. I tested a specification that allowed received citations to influence both aspects of the distribution, but found that they had no significant influence on σ .¹⁹

Finally, the renewal data on highly cited patents suggest that a large portion are not very valuable. Of the top 10% of patents ranked by citations in 1991 (with 15 or more citations received), 37% were not renewed to term. Among the top 5% (with 21 or more citations received), 32% were not renewed to term. Thus even among the most highly cited patents, a large portion have low values. This means that citations and patent value cannot be closely correlated overall, although there might be a stronger relationship among a subset of highly cited patents. Nevertheless, these data suggest that highly cited patents in the upper tail are not likely to cause my estimates of the effect of patent citations on patent value to be seriously understated.

This analysis emphasizes not only that patent citations are “noisy,” but also that care must be taken in interpreting the meaning of correlations involving citation statistics. Patent citations may be a good (but noisy) indicator of technology value, but they do not appear to be closely related to patent value. For this reason, they might not be reliable as a measure of patent “quality.”²⁰

3.3. Estimates for public manufacturing firms

Table 6 reports regressions for the panel of patents granted to public firms from 1985 through 1991. It is important to check estimates over a range of cohorts, to make sure that temporary disequilibria do not affect the estimates. Column 1 shows a simple regression, comparable to those used in the 1991 sample. The coefficients and the value estimates are quite similar to those obtained for large entities in 1991 in Table 2 and for publicly listed firms in 1991 in Table 3. This provides some assurance that temporary effects do not appear to exert much effect on the estimates.

It is also possible that patent values may have changed over time. To test this, I ran the regression in column 1 with dummy variables for each grant year and then again with dummies for each application year. These dummy coefficients are shown in Fig. 1 (normalized to equal 1 in 1985). These suggest that patent value increased modestly during the mid-1980s and then leveled off. This is consistent with the notion that the creation of a centralized appeals court for patents in 1982 (the Court of Appeals for the Federal Circuit) may have “strengthened” patents, thus increasing their value. This is also roughly contemporaneous with the well-known acceleration in patenting rates, which has been a subject of several studies (Kortum and Lerner, 1999; Hall and Ziedonis, 2001; Henry and Turner, 2006; Sanyal, 2005). Unfortunately, estimates of patent value for years before 1983 are unreliable because of sample size and data quality problems.

The remaining columns of Table 6 explore firm characteristics. In column 2, patent value increases with firm R&D spending. This is consistent with the view that more valuable patents are correlated with more valuable technology. However, the larger the patent stock of the firm, all

²⁰ Patent citations might still play a role in more complicated models of patent value (see, for example, Leiva, 2007), but my results indicate that patent citations do not appear to be a reliable *index measure* of patent value, which is how some researchers have used them.

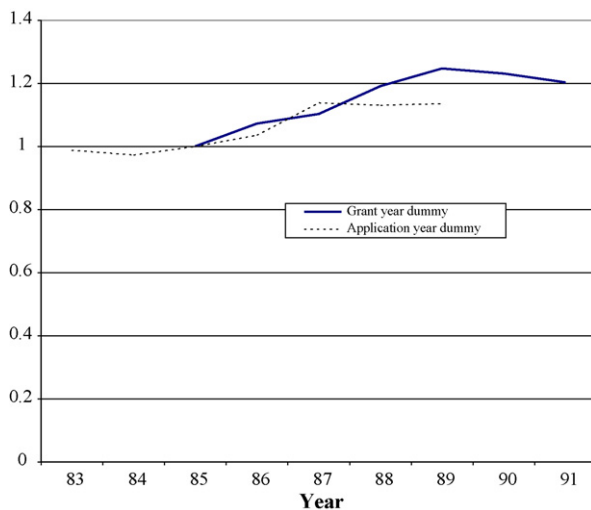
¹⁹ The coefficient and standard error are 0.0002 (0.0012).

Table 6

Regressions with firm characteristics for U.S. public manufacturing firms, 1985–1991

	1	2	3
Small entity	−0.93 (0.05)	−0.99 (0.05)	−0.99 (0.05)
New firm		−0.08 (0.03)	
Ln R&D		0.21 (0.01)	
Ln employment		0.01 (0.01)	
Ln patent stock		−0.29 (0.01)	
Industry			
Pharmaceuticals			0.35 (0.03)
Electrical and instrument			0.34 (0.02)
Electronic and computer			0.85 (0.02)
Metals and transportation			0.08 (0.02)
Constant	8.69 (0.06)	9.04 (0.07)	8.40 (0.05)
σ	1.87 (0.03)	1.85 (0.03)	1.84 (0.03)
d	0.16 (0.00)	0.16 (0.00)	0.16 (0.00)
Median net present value	18010	17934	17786
Mean net present value	113066	112695	110976
Number of observations	107364	107364	107364
Log L	−133549.7	−132876.7	−132387.2

Note: Robust standard errors in parentheses. The new firm dummy is one if the firm has been publicly listed for fewer than 5 years. R&D is deflated R&D stock, employment is in thousands, and patent stock is calculated using the perpetual inventory methods at a 15% depreciation. Patent values in 1992 \$, discounted at 10%.

**Fig. 1.** Year effects. Note: Series are normalized to equal 1 in 1985.

else equal, the *smaller* the mean patent value. This suggests that there are diminishing returns to patenting—patent values decrease as more patents are obtained, all else equal. Column 2 also includes a dummy variable for “new” firms,

that is, firms that have been publicly listed for fewer than 5 years. It is sometimes argued that patents are particularly valuable to new firms. Although patents might be valuable relative to alternatives, this regression finds, to the contrary, that new firms have patents with significantly *smaller* value.

Column 3 reports industry dummies. Computer and electronics industries have the largest patent value, and “other” (the omitted category) has the lowest.

4. Analysis

4.1. Comparisons to other estimates

Table 7 compares my estimates to those obtained by other researchers for U.S. patents. Based on the above analysis, we should expect substantial differences depending on the particular population of patents being compared. Barney (2002) uses the 1986 cohort of patents granted and obtains slightly smaller estimates. Serrano (2005) estimates of patent value using both renewal and re-assignment data for a group of U.S. organizations that do not patent heavily. In 1992 dollars he obtains a mean value of \$48,000 and a median value of \$17,000. These are smaller,

Table 7

Comparison of estimates of patent value

Study	Patent issue year	Patent country	Group	Patent value (92 U.S.)	
				Median	Mean
This paper	1991	U.S.	All U.S. patentees	7,175	78,168
	1985–1991	U.S.	U.S. public firms, manufacturing	18,010	113,067
Other U.S. studies					
Barney (2002)	1986	U.S.	All	5,849	61,896
Serrano (2005)	1983–2002	U.S.	Small business patentees	17,361	47,456
Putnam (1996)	1974	U.S.	Also filed abroad		188,355
			All (imputed)		78,800

but Serrano's sample is likely weighted more toward small firms, who, as we have seen, have patents with smaller values.

Putnam (1996) uses a sample of patents that were also filed in one or more other countries. Using data on international filings (in a model similar to the renewal model), he estimates that patents that were successfully filed in the U.S. in 1974 that were also filed abroad were worth \$188,000 in 1992 dollars. In general, patents that are filed in multiple countries tend to be much more valuable than patents that are not, so it is not surprising that Putnam's mean estimate is substantially higher than the others. Using Putnam's data it is possible to impute the mean value of all U.S. patents, including those that were only filed domestically. This figure is about \$79,000.²¹ Thus all of these estimates of U.S. mean patent value are roughly consistent.

The estimated value of European patents is also substantially smaller than the U.S. estimates. Converted to 1992 U.S. dollars, the mean values from five studies range from about \$2000 to \$32,000, with an average of about \$16,000.²² However, it is not surprising that these values are so much larger because the U.S. market is much larger than any of the national European markets.

As discussed above, one concern about estimates of patent value based on renewal data is that patentee renewal decisions do not directly reveal the values of the most valuable patents in the upper tail. This means that although estimates of median value are based on an observed distribution, estimates of mean patent value are based on an extrapolation. In a separate paper (Bessen, 2006a), I address the valuation of the upper tail by reviewing estimates of patent value obtained from analysis of firm market value. This analysis suggests that the estimates based on renewal data do not substantially understate patent value for these firms, except, possibly, for large pharmaceutical firms.

4.2. Estimates of the patent subsidy

Patent rents represent the "reward" that patents afford inventors. Schankerman (1998) suggests that this reward can be considered as equivalent to an R&D subsidy. He asks what subsidy would be needed in order to induce firms to make the same investment in R&D as they are induced to make by patents. He suggests that the ratio of a patentee's aggregate patent value (=present value of patent rents)

divided by the associated R&D expenditure can be considered an upper bound estimate of this "equivalent subsidy rate."²³

Several researchers have calculated equivalent subsidy rates by multiplying estimates of patent value (from European renewal studies) times the ratio of successful patent applications to dollars of R&D (Pakes, 1986; Pakes and Schankerman, 1984; Lanjouw, 1998; Schankerman, 1998). These studies obtain estimates for patent cohorts from the 1970s that range from 4% to 35%, averaging about 18%.²⁴

However, these ratios were calculated using aggregate national data (aggregate patent counts and aggregate R&D) pro-rated to the country where the patent was granted (e.g., R&D performed in the U.S., Germany, etc. is allocated to French patents). Unfortunately, estimates derived from aggregate data from the 1970s might not accurately represent the subsidy that a modern firm can expect to get on its R&D investment.²⁵

Firm-level data permit a more representative calculation that compares the value of the patents a firm obtains in the U.S. to the level of that firm's R&D. Table 8 shows a variety of estimates calculated at different levels of aggregation. The first two rows display aggregate data for all U.S. patentees (foreign organizations and governments excluded), using the estimated value per patent from Table 2, column 1. The aggregate value of the 56,957 patents granted in 1991 to U.S. patentees was about \$4.4 billion in 1992 dollars. I calculated the corresponding R&D investment using data from the National Science Foundation (NSF) survey of U.S. firms. Since the patents granted in 1991 were applied for over many years, column 4 displays a weighted sum of real R&D spending where the weights are allocated based on the proportion of 1991 patents applied for in each year R&D was performed. Column 6 displays the equivalent subsidy rate. The first row shows the calculation using all industrial R&D, the second row shows the rate using just company-funded R&D. These estimates are within the range of some of the estimates based on European data, but at the bottom of that range.

The remaining rows reports estimates using data on firm R&D spending for publicly listed firms in the Compustat database. These estimates are conditional on firms' choosing to patent (about 15% of R&D is performed by firms that do not patent). The third row reports the calculation for all publicly listed firms matched to patent data in 1991. The R&D figure in column 4 is a weighted sum of each firm's R&D expenditure for the application year for each patent granted

²¹ Putnam (1996) reports that in 1974, 36% of US patents were also filed in another country (Table 3.3). Patents granted in the US that were also filed abroad were worth \$188,000 in 1992 dollars. Putnam also estimates that in aggregate, domestic patents (estimated for Germany) add about 5% to the aggregate value of all international patents held in a country (p. 129). Worldwide rights associated with each US international patent was \$609,600 in 1974. Then the mean value of all patents should be $(\$188,355 + 0.05 \times \$609,600) \times \text{no. of int'l patents} / \text{total no. of patents} = (\$188,355 + 0.05 \times \$609,600) \times 0.36 = \$78,800$. Alternatively, if the domestic-only patents are assumed to be worth \$20,000 (according to patent attorneys, the cost of filing in one additional English-speaking country is under \$20,000), then the mean patent value is about \$80,000.

²² These include Baudry and Dumont (2006), Lanjouw (1998), Schankerman (1998), Pakes (1986) and Pakes and Schankerman (1984). See Bessen (2006b) for more details.

²³ Schankerman (1998) and Lanjouw (1998) point out that this may be an upper bound estimate for several reasons. There may be diminishing returns to R&D. A cash subsidy may be relatively more valuable to risk averse investors, so they might require a smaller equivalent subsidy. Also, this analysis does not take into account strategic interaction; to the extent each firm's patents reduce the rents that other firms earn, the subsidy needed in a world without patents would also be less.

²⁴ A summary of these results is shown in Table 8 of Bessen (2006b).

²⁵ There are several reasons why estimates based on aggregate data may be misleading: these estimates include patents of individuals and small inventors who are not included in the R&D data, the R&D data used was likely under-reported, and the methods used to pro-rate R&D spending produce a measure of the returns to worldwide patents, not just the returns to patents in the subject country. See Bessen (2006b) for details.

Table 8
Equivalent subsidy rate

1. Patentees	2. Aggregate patent value (mill. \$92)	3. R&D source (allocated over patent application years)	4. Deflated R&D (mill. \$92)	5. Patents/\$ mill. R&D	6. Equivalent subsidy rate (%)
1991 sample					
All U.S. patentees	\$4441	NSF—total U.S. R&D	\$115,892	0.49	3.8
All U.S. patentees	\$4441	NSF—U.S. Company R&D	\$84,291	0.93	5.3
Publicly listed U.S. firms	\$1989	Firm R&D—Compustat	\$69,752	0.31	2.9
1985–1991 panel					
Publicly listed U.S. manufacturing firms	\$12,758	Firm R&D—Compustat	\$446,597	0.25	2.9

Note: Estimates for firms are conditional on firms choosing to patent.

in 1991, the weights apportioning each year's R&D equally across all patents for that year. The patent value comes from Table 3. Even though the value per patent is higher, the estimate of the equivalent subsidy rate is much smaller, 2.9%. The main reason for this is that the ratio of patents to R&D is much lower (see column 5) because so much more of measured R&D is performed by public firms. The fourth row repeats this calculation using the sample of patents from publicly listed manufacturing firms for 1985–1991, using the mean patent value estimates from Table 6, column 1. This estimate of the equivalent subsidy rate is also 2.9%.

Thus firm-level data suggests that the equivalent subsidy provided by U.S. patents to public U.S. firms is about 3%. Note that this is the subsidy provided *only* by firms' U.S. patents—they will earn additional subsidies from patents obtained on the same inventions in other countries. Thumbail calculations (see footnote 21) suggest that the total subsidy provided by worldwide patents may be two or three times larger or more, somewhat more consistent with the concept measured in the studies by Lanjouw, Pakes and Schankerman.

Arora et al. (2003) also estimate an equivalent subsidy ratio using a structural model incorporating survey data and firm-level R&D data. Their estimate of 17% also represents the return on worldwide patent rights, so it, too, is not directly comparable.

5. Conclusion

I have used U.S. patent renewal data to estimate the value of U.S. patents to different groups of domestic patentees and to analyze some of the determinants of patent value. Among my findings

- I estimate that patents granted to U.S. patentees in 1991 were worth about \$78,000 in the mean (\$7000 in the median) to their owners. This is substantially larger than estimates made using similar data for European patents. Estimates for a panel of patents granted to public manufacturing firms were higher, about \$113,000 in the mean and \$18,000 in the median.
- Despite this larger per patent value, I find that the ratio of patent value to R&D – a measure of the subsidy that patents provide to R&D investment – is only about

3%. Nevertheless, in aggregate, the value of U.S. patents granted to U.S. owners in 1991 was over \$4 billion. This is large, but it is substantially smaller than other government programs to encourage R&D. For example, direct Federal funding of industrial R&D in 1991 was over \$26 billion.

- There are large differences in patent value across different groups of patentees. Small entities – individuals, corporations with fewer than 500 employees and non-profit organizations – have patent values that are on average less than half as large as the values obtained by large corporations. Contrary to lore, patents do not seem to work particularly well for small inventors, although they might nevertheless be important to small inventors. Some evidence suggests that this disparity disappears in technology fields where markets for technology work particularly well.
- I quantify the association between litigation and patent citation statistics and patent value. A litigated patent is worth nearly six times as much as a non-litigated patent. An additional patent citation received increases patent value by 4–7%. Although patent citations are significantly correlated with patent value, they only explain a small portion of the variance in patent value, so they are not reliable measures of patent quality. Patent citation statistics may be more meaningful as measures of the value of the underlying technology.

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References

- Allison, J.R., Lemley, M., Moore, K.A., Trunkey, D., 2004. Valuable patents. 92 Georgetown Law Journal, 435.
- Arora, A., Ceccagnoli, M., Cohen, W.M., 2003. R&D and the Patent Premium. NBER Working Paper No. 9431.
- Barney, J.A., 2002. A study of patent mortality rates: using statistical survival analysis to rate and value patent assets. AIPLA Quarterly Journal 30 (3), 317–352.

- Baudry, M., Dumont, B., 2006. Patent renewals as options: improving the mechanism for weeding out lousy patents. *Review of Industrial Organization* 28 (1), 41–62.
- Bessen, J., 2006a. Estimates of Firms' Patent Rents from Firm Market Value. Boston University School of Law Working Paper No. 06-14.
- Bessen, J., 2006b. The Value of U.S. Patents by Owner and Patent Characteristics. Boston University School of Law Working Paper No. 06-46.
- Bessen, J., Meurer, M.J., 2005. The Patent Litigation Explosion. Working Paper.
- Bessen, J., Meurer, M.J., 2007. The Private Cost of Patent Litigation. Boston University School of Law Working Paper No. 07-08.
- Bessen, J., Hunt, R.M., 2004. An Empirical Look at Software Patents. Working Paper No. 03-17/R. Federal Reserve Bank of Philadelphia.
- Deng, Y., 2005. A Dynamic Stochastic Analysis of International Patent Application and Renewal Processes. Working Paper.
- Gustafsson, C., 2005. Private Value of Patents in a Small Economy—Evidence From Finland. Working Paper.
- Hall, B.H., Jaffe, A.B., Trajtenberg, M., 2001. The NBER Patent Citations Data File: Lessons, Insights and Methodological Tools. NBER Working Paper No. 8498.
- Hall, B., Jaffe, A., Trajtenberg, M., 2005. Market value and patent citations. *Rand Journal of Economics*, 36.
- Hall, B.H., Ziedonis, R., 2001. The patent paradox revisited: an empirical study of patenting in the U.S. semiconductor industry, 1979–1995. *RAND Journal of Economics* 32, 101–128.
- Harhoff, D., Scherer, F.M., Vopel, K., 2003a. Exploring the tail of patented invention value distributions. In: Ove, G. (Ed.), *Economics, Law and Intellectual Property*. Kluwer Academic Publishers, The Netherlands, pp. 279–309.
- Harhoff, D., Scherer, F.M., Vopel, K., 2003b. Citations, family size, opposition and the value of patent rights. *Research Policy* 32 (8), 1343–1363.
- Harhoff, D., Francis, N., Scherer, F.M., Vopel, K., 1999. Citation frequency and the value of patented inventions. *Review of Economics and Statistics* 81 (3), 511–515.
- Henry, M., Turner, J.L., 2006. The Court of Appeals for the Federal Circuit's Impact on Patent Litigation. *Journal of Legal Studies* 35, 85–117.
- Kortum, S., Lerner, J., 1999. What is behind the recent surge in patenting? *Research Policy* 28 (1), 1–22.
- Lanjouw, J.O., 1998. Patent protection in the shadow of infringement: simulation estimations of patent value. *Review of Economic Studies* 65, 671–710.
- Lanjouw, J.O., Pakes, A., Putnam, J., 1998. How to count patents and value intellectual property: the uses of patent renewal and application data. *Journal of Industrial Economics* 46, 405–432.
- Lanjouw, J.O., Schankerman, M., 2004a. Patent quality and research productivity: measuring innovation with multiple indicators. *Economic Journal* 114 (495), 441–465.
- Lanjouw, J.O., Schankerman, M., 2004b. Protecting intellectual property: are small firms handicapped? *Journal of Law and Economics* XLVII, 45–74.
- Lehman, B.A., 1993. Annual Report Fiscal 1993. U.S. Patent and Trademark Office.
- Leiva, F., 2007. Innovations as a Network of Ideas: Assessing Patent Values Through Citations. Working Paper.
- Marco, A.C., 2005. The option value of patent litigation: theory and evidence. *Review of Financial Economics* 14, 323–351.
- Moore, K.A., 2005. Worthless Patents. George Mason Law & Economics Research Paper No. 04-29.
- National Science Foundation, 2008. Industrial Research and Development Information System. <http://www.nsf.gov/statistics/iris/>.
- Pakes, A., 1986. Patents as options: some estimates of the value of holding European patent stocks. *Econometrica* 54, 755–784.
- Pakes, A., Schankerman, M., 1984. The rate of obsolescence of patents, research gestation lags, and the private rate of return to research resources. In: Griliches, Z. (Ed.), *R&D, Patents and Productivity*. University of Chicago Press for the NBER, Chicago.
- Putnam, J., 1996. The Value of International Patent Protection. PhD Thesis. Yale.
- Sanyal, P., 2005. The Patent Explosions: Quantifying Changes in the Propensity to Patent. Brandeis University Working Paper.
- Schankerman, M., 1998. How valuable is patent protection: estimates by technology fields. *RAND Journal of Economics* 29 (1), 77–107.
- Schankerman, M., Pakes, A., 1986. Estimates of the value of patent rights in European countries during the post-1950 period. *The Economic Journal* 96, 1052–1076.
- Serrano, C.J., 2005. The Market for Intellectual Property: Evidence From the Transfer of Patents. Working Paper.
- Trajtenberg, M., 1990. A penny for your quotes: patent citations and the value of innovations. *RAND Journal of Economics* 21 (1), 172–187.
- Trajtenberg, M., Henderson, R., Jaffe, A., 1997. University versus corporate patents: a window on the basicness of invention. *Economics of Innovation and New Technology* 5, 19–50.